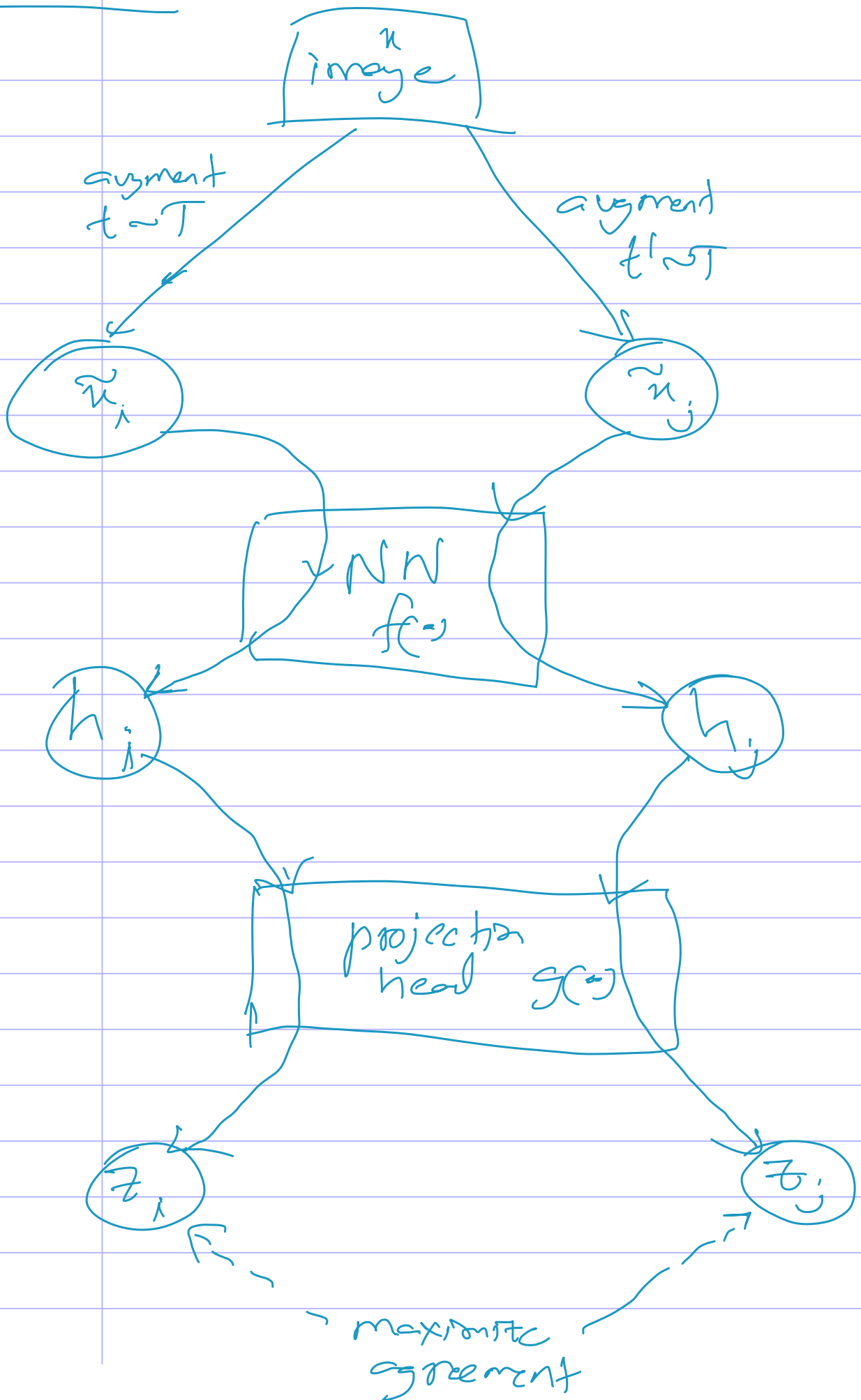


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① SimCLR



cosine similarity

$$\text{sim}(\underline{u}, \underline{v}) = \frac{\underline{u}^T \underline{v}}{\|\underline{u}\| \|\underline{v}\|}$$

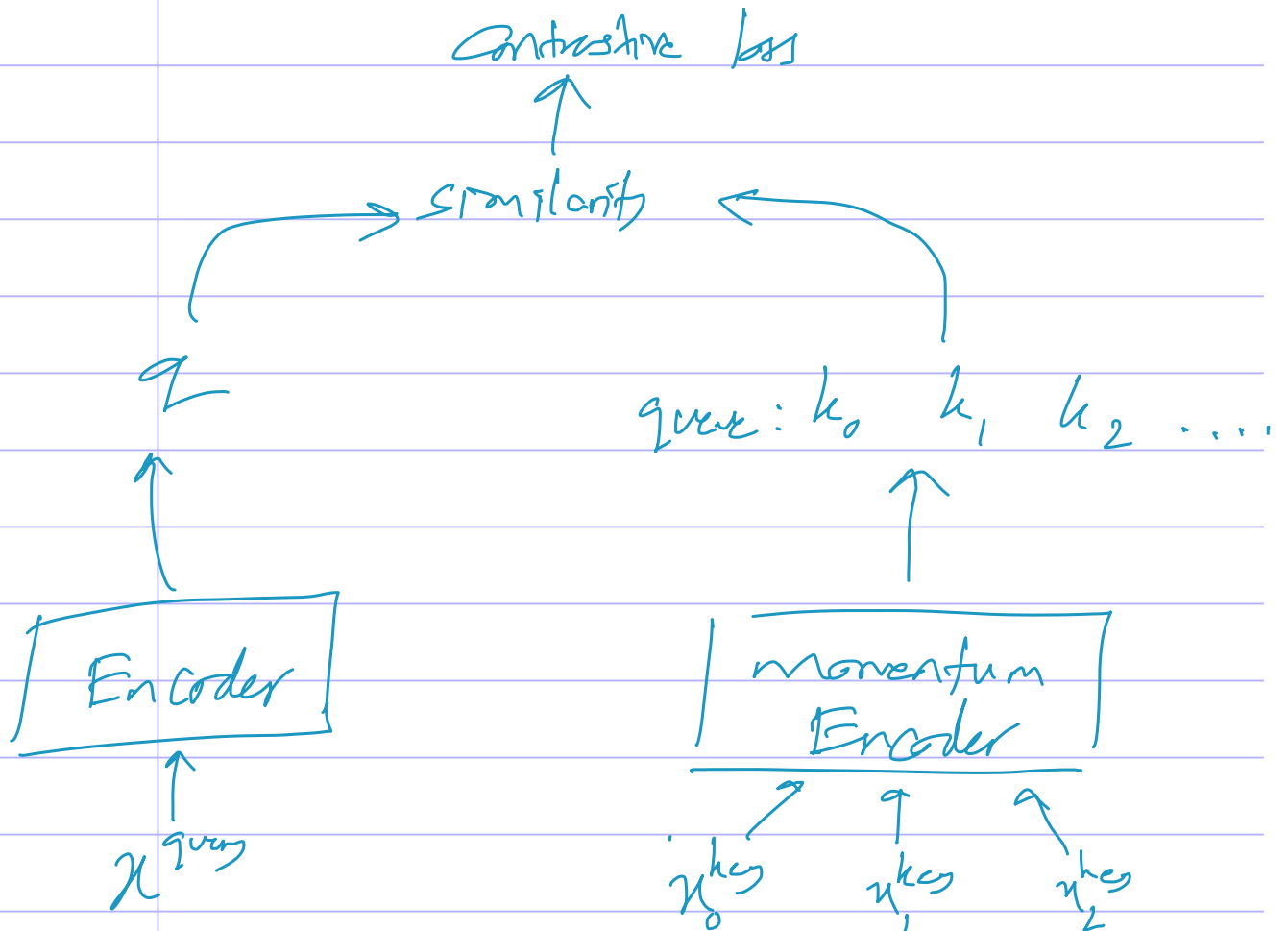
log funcⁿ for positive pair (i, j)

$$l_{ij} = -\log \left(\frac{\exp[\text{sim}(\underline{z}_i, \underline{z}_j) / t]}{\sum_{k=1}^{2N} \exp[\text{sim}(\underline{z}_i, \underline{z}_k) / t]} \right)$$

$k \neq i$

② MoCo

6/4/2026



Info NCE loss:

$$L_q = -\log \left[\frac{\exp(q \cdot k_+ / \tau)}{\sum_{i=1}^K \exp(q \cdot k_i / \tau)} \right]$$